QuakeNet

Earthquake Early Warning and Emergency Response Network — Design and Implementation of a National Disaster-Response Network Using VLSM, RIPv2, Static Routing, Recursive and Floating Backup Routes

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Project type	Networking course project (Cisco Packet Tracer), Summer 2026
Base network	18.50.0.0/16 (derived from the last four digits, 1850, of the student ID assigned by the project specification)
Administrative distance	48 (derived from the digit sum of an assigned student ID × group size)
Contents of this repository	Packet Tracer file, router/switch configurations, addressing tables, services, automated design verifier

1. Introduction and Objectives

QuakeNet is a simulated national Earthquake Early Warning and Emergency Response Network built in Cisco Packet Tracer for a Ministry of Disaster Management. The network connects seven operational units: the National Emergency Operation Center (NEOC), the Seismic Monitoring Station (SMS), the Fire and Rescue Headquarters (FRH), the Medical Emergency Center (MEC), the Urban Search and Rescue Camp (USR), the Communication Relay Center (CRC) and the Mountain Observation Post (MOP).

Each unit is represented by a dedicated router, a dedicated LAN switch, two PCs that represent all the users of that unit, and one network printer. Every unit also hosts its own email server. The NEOC is the core of the network: it hosts the central DNS server, one of the two web servers, and serves DHCP for three units. The MEC is a second service centre, hosting the other web server and its own DHCP server.

The design deliberately mixes routing styles. The three units on the warning loop run RIPv2, while the remaining four are reached exclusively through static routes in three styles: recursive next-hop routes, exit-interface routes and floating backup routes. **No default route is permitted anywhere in the system**, so every destination must be reachable through a specific network route.

The main objectives are:

- Derive a base network from a student ID and subnet it efficiently with VLSM for seven LANs of widely differing sizes (50 to 5000 hosts).
- Design a topology that satisfies every connectivity constraint of the specification, including a shared central-switch segment, a point-to-point fibre link and a three-router warning loop.
- Combine RIPv2 with several styles of static routing, and redistribute static routes into RIPv2 so both halves of the network learn each other.
- Configure centralised DHCP on a router, distributed DHCP on dedicated servers, and DHCP relay with `ip helper-address` for remote units.
- Configure a central DNS service, two web servers and seven email servers, and demonstrate cross-domain email between two independent mail domains.
- Implement a recursive backup route and a floating static route (with IP SLA tracking) so a backup path is used automatically when a primary link fails, and verify this with two independent link-failure tests.

2. Assumptions

Each assumption is technically sound, does not conflict with any requirement, and does not remove or simplify any compulsory feature.

1. **Users.** Two PCs and one printer represent all the users of each unit. Every VLSM block is still sized for the full stated host count (5000, 1040, 500, 320, 180, 100 and 50).
2. **Hardware.** Cisco ISR4331 routers for all seven units; Cisco 2960-24TT switches for all eight switches (seven unit switches and one central switch).
3. **Serial modules.** A NIM-2T module is installed in every router to provide the serial WAN interfaces of the warning loop and the two stub links.
4. **Clocking.** All serial links run at 64000 bps, clock rate set on the DCE end; interface bandwidth is left at its default.
5. **Static addressing.** Servers, printers and switch management interfaces (VLAN 1) use static IP addresses. Only the

unit PCs use DHCP.

6. **Fibre link.** The NEOC–MEC fibre link is implemented as a GigabitEthernet point-to-point connection, because the ISR4331 has no fibre-capable port without extra modules. No routing requirement changes.
7. **Central segment.** The four routers on the central switch share one broadcast domain, addressed as a /29 — the smallest block for four router interfaces plus the switch management address.
8. **USR router.** It uses directly attached (exit-interface only) static routes, valid because its single uplink is a point-to-point serial link.
9. **Return-path backups on NEOC.** Floating routes with administrative distance 48 are also configured on NEOC towards the MEC and USR LANs. Without them the traffic would fail in the return direction during a fibre failure.
10. **Backup distance.** Administrative distance 48 is used for every backup route.
11. **IP SLA tracking.** IP SLA with object tracking is configured on CRC so the floating static route can genuinely be demonstrated: both of CRC's candidate next hops lie on the same connected /29, so without tracking the primary route would never be withdrawn.
12. **No default route** (0.0.0.0/0) exists anywhere, as required.
13. **Credentials.** `cisco123` for all email accounts; `cisco` / `class` for console and privileged EXEC access — lab-only values, chosen for simplicity.
14. **Mail domains.** Each unit uses `<unit>.gov`. Only NEOC and MEC need two mailboxes; the other five units have one demonstration mailbox each.

3. Base Network and Administrative Distance

3.1 Base network address

The base network is derived from the student ID that the project specification designates; only its last four digits are needed:

Step	Operation	Result
1	Take the last four digits of the designated student ID	1850
2	Split into two groups of two digits	18 and 50
3	First pair becomes the first octet	18
4	Second pair becomes the second octet	50
5	Third and fourth octets are zero, prefix is /16	**18.50.0.0/16**

The base network is **18.50.0.0/16** (default mask 255.255.0.0). This /16 provides 65,534 usable host addresses, far more than the 7,190 users of the seven units; the surplus is deliberately left unallocated for future units.

3.2 Administrative distance

The administrative distance for every backup route is derived from a second assigned student ID, multiplied by the group size:

Step	Operation	Result
1	Sum of the digits of the assigned student ID	12
2	Number of members in the group	4
3	Administrative distance = digit sum × group size	12 × 4 = **48**

An administrative distance of 48 is higher than the default of 1 for a static route, so a backup route with AD 48 stays out of the routing table while the primary static route is valid and installs automatically the moment the primary is withdrawn. (It is also lower than RIP's 120, which matters for floating routes that compete with RIP-learned prefixes.)

4. VLSM Design and Subnetting Table

VLSM sorts all required networks from the largest host requirement to the smallest and allocates aligned blocks sequentially from the start of 18.50.0.0/16, which prevents fragmentation and guarantees no overlap. The allocation is reproduced by `tools/vlsm.py` and checked against the table below by the automated verifier.

4.1 Host requirement analysis

Network	Required hosts	Block size	Prefix	Usable hosts
NEOC LAN	5000	8192	/19	8190

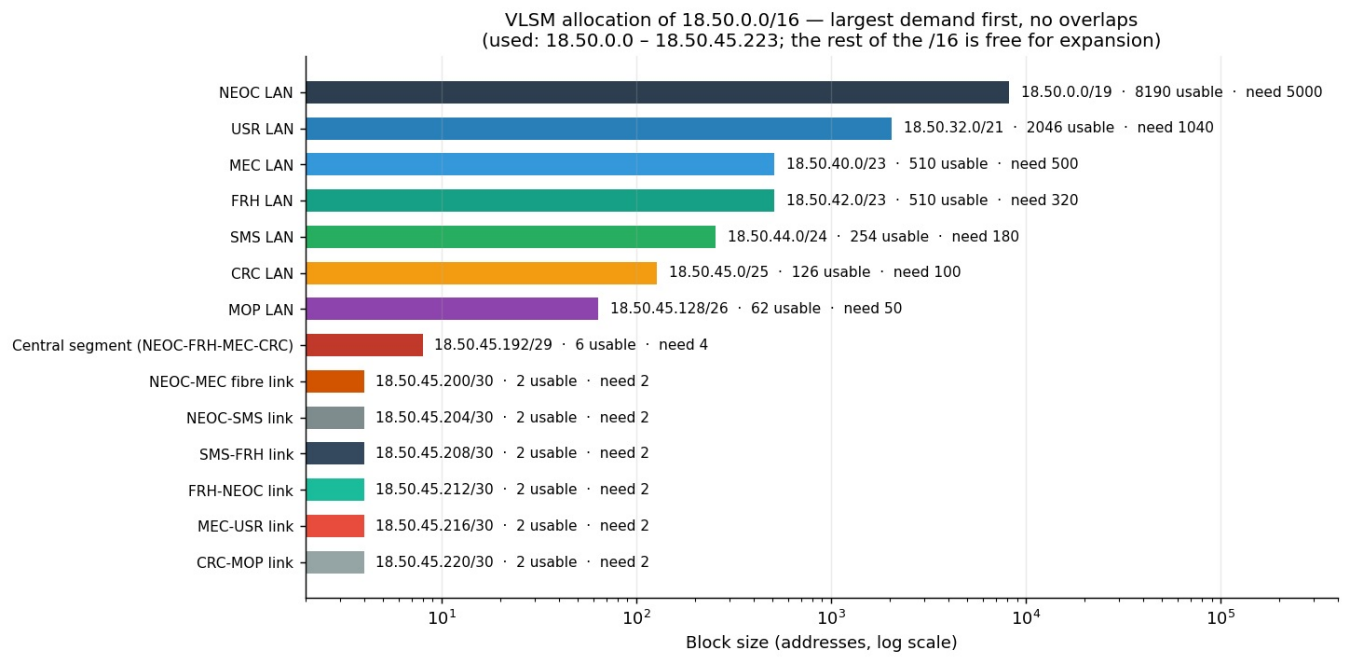
NEOC LAN	5000	8192	/19	8190
USR LAN	1040	2048	/21	2046
MEC LAN	500	512	/23	510
FRH LAN	320	512	/23	510
SMS LAN	180	256	/24	254
CRC LAN	100	128	/25	126
MOP LAN	50	64	/26	62
Central segment (NEOC, FRH, MEC, CRC)	4	8	/29	6
NEOC ↔ MEC fibre link	2	4	/30	2
NEOC ↔ SMS link	2	4	/30	2
SMS ↔ FRH link	2	4	/30	2
FRH ↔ NEOC link	2	4	/30	2
MEC ↔ USR link	2	4	/30	2
CRC ↔ MOP link	2	4	/30	2

The NEOC LAN needs 5000 hosts. A /20 provides only 4094 usable addresses, so a /19 (8190) was allocated. Likewise the USR LAN needs 1040 hosts, so a /21 (2046) was allocated rather than a /22, which provides only 1022.

4.2 Complete subnetting table

#	Network	Subnet	Mask	Usable range	Broadcast
1	NEOC LAN	18.50.0.0/19	255.255.224.0	18.50.0.1 – 18.50.31.254	18.50.31.255
2	USR LAN	18.50.32.0/21	255.255.248.0	18.50.32.1 – 18.50.39.254	18.50.39.255
3	MEC LAN	18.50.40.0/23	255.255.254.0	18.50.40.1 – 18.50.41.254	18.50.41.255
4	FRH LAN	18.50.42.0/23	255.255.254.0	18.50.42.1 – 18.50.43.254	18.50.43.255
5	SMS LAN	18.50.44.0/24	255.255.255.0	18.50.44.1 – 18.50.44.254	18.50.44.255
6	CRC LAN	18.50.45.0/25	255.255.255.128	18.50.45.1 – 18.50.45.126	18.50.45.127
7	MOP LAN	18.50.45.128/26	255.255.255.192	18.50.45.129 – 18.50.45.190	18.50.45.191
8	Central segment	18.50.45.192/29	255.255.255.248	18.50.45.193 – 18.50.45.198	18.50.45.199
9	NEOC ↔ MEC	18.50.45.200/30	255.255.255.252	18.50.45.201 – 18.50.45.202	18.50.45.203
10	NEOC ↔ SMS	18.50.45.204/30	255.255.255.252	18.50.45.205 – 18.50.45.206	18.50.45.207
11	SMS ↔ FRH	18.50.45.208/30	255.255.255.252	18.50.45.209 – 18.50.45.210	18.50.45.211
12	FRH ↔ NEOC	18.50.45.212/30	255.255.255.252	18.50.45.213 – 18.50.45.214	18.50.45.215
13	MEC ↔ USR	18.50.45.216/30	255.255.255.252	18.50.45.217 – 18.50.45.218	18.50.45.219
14	CRC ↔ MOP	18.50.45.220/30	255.255.255.252	18.50.45.221 – 18.50.45.222	18.50.45.223

The address space from 18.50.45.224 to 18.50.255.255 remains unallocated and is reserved for future expansion.

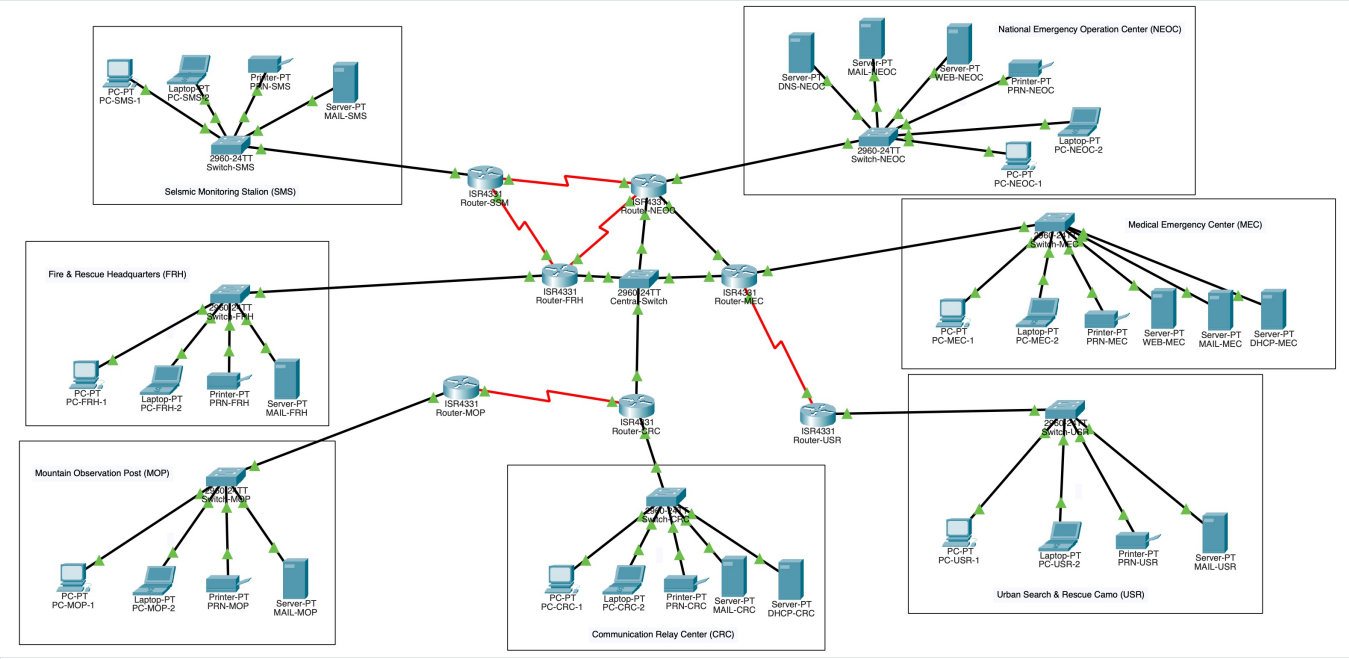


VLSM allocation of 18.50.0.0/16

5. Network Topology

The topology satisfies every structural constraint of the specification:

- Each of the seven operational units is represented by a separate router.
- MOP is connected only to CRC, through a serial link on 18.50.45.220/30.
- NEOC, FRH, MEC and CRC are connected through one central switch on the 18.50.45.192/29 segment.
- A direct point-to-point fibre link joins NEOC and MEC on 18.50.45.200/30, in addition to their shared central-switch connection.
- The warning loop is formed by three links: NEOC ↔ SMS, SMS ↔ FRH and FRH ↔ NEOC.
- USR is connected only to MEC, through a serial link on 18.50.45.216/30.
- Each unit has its own LAN switch, two PCs and one printer.
- Each unit hosts its own email server; only NEOC and MEC host web servers; only NEOC hosts the central DNS server.



QuakeNet labelled topology

5.1 Physical connections

#	From	Interface	To	Interface	Link type

1	Router-NEOC	G0/0/0	Switch-NEOC	Fa0/1	Copper straight-through
2	Router-NEOC	G0/0/1	Central-Switch	Fa0/1	Copper straight-through
3	Router-FRH	G0/0/1	Central-Switch	Fa0/2	Copper straight-through
4	Router-MEC	G0/0/1	Central-Switch	Fa0/3	Copper straight-through
5	Router-CRC	G0/0/1	Central-Switch	Fa0/4	Copper straight-through
6	Router-NEOC	G0/0/2	Router-MEC	G0/0/2	Point-to-point fibre link
7	Router-NEOC (DCE)	S0/1/0	Router-SMS	S0/1/0	Serial, 64000 bps
8	Router-SMS (DCE)	S0/1/1	Router-FRH	S0/1/0	Serial, 64000 bps
9	Router-NEOC (DCE)	S0/1/1	Router-FRH	S0/1/1	Serial, 64000 bps
10	Router-MEC (DCE)	S0/1/0	Router-USR	S0/1/0	Serial, 64000 bps
11	Router-CRC (DCE)	S0/1/0	Router-MOP	S0/1/0	Serial, 64000 bps
12	Each unit router	G0/0/0	Its unit switch	Fa0/1	Copper straight-through
13	PCs, printers, servers	Fa0	Unit switch	Fa0/2 onward	Copper straight-through

6. Device IP Addressing

6.1 Router interfaces

Router	Interface	IP address	Subnet mask	Connects to
Router-NEOC	G0/0/0	18.50.0.1	255.255.224.0	Switch-NEOC
	G0/0/1	18.50.45.193	255.255.255.248	Central-Switch
	G0/0/2	18.50.45.201	255.255.255.252	Router-MEC (fibre)
	S0/1/0 (DCE)	18.50.45.205	255.255.255.252	Router-SMS
	S0/1/1 (DCE)	18.50.45.214	255.255.255.252	Router-FRH
Router-SMS	G0/0/0	18.50.44.1	255.255.255.0	Switch-SMS
	S0/1/0 (DTE)	18.50.45.206	255.255.255.252	Router-NEOC
	S0/1/1 (DCE)	18.50.45.209	255.255.255.252	Router-FRH
Router-FRH	G0/0/0	18.50.42.1	255.255.254.0	Switch-FRH
	G0/0/1	18.50.45.194	255.255.255.248	Central-Switch
	S0/1/0 (DTE)	18.50.45.210	255.255.255.252	Router-SMS
	S0/1/1 (DTE)	18.50.45.213	255.255.255.252	Router-NEOC
Router-MEC	G0/0/0	18.50.40.1	255.255.254.0	Switch-MEC
	G0/0/1	18.50.45.195	255.255.255.248	Central-Switch
	G0/0/2	18.50.45.202	255.255.255.252	Router-NEOC (fibre)
	S0/1/0 (DCE)	18.50.45.217	255.255.255.252	Router-USR
Router-CRC	G0/0/0	18.50.45.1	255.255.255.128	Switch-CRC
	G0/0/1	18.50.45.196	255.255.255.248	Central-Switch
	S0/1/0 (DCE)	18.50.45.221	255.255.255.252	Router-MOP
Router-USR	G0/0/0	18.50.32.1	255.255.248.0	Switch-USR
	S0/1/0 (DTE)	18.50.45.218	255.255.255.252	Router-MEC
Router-MOP	G0/0/0	18.50.45.129	255.255.255.192	Switch-MOP
	S0/1/0 (DTE)	18.50.45.222	255.255.255.252	Router-CRC

6.2 Servers, printers and switches

Device	Role	IP address	Subnet mask	Default gateway	DNS

DNS-NEOC	Central DNS	18.50.0.2	255.255.224.0	18.50.0.1	18.50.0.2
WEB-NEOC	www.neoc.alert.gov	18.50.0.3	255.255.224.0	18.50.0.1	18.50.0.2
MAIL-NEOC	neoc.gov	18.50.0.4	255.255.224.0	18.50.0.1	18.50.0.2
PRN-NEOC	Printer	18.50.0.5	255.255.224.0	18.50.0.1	18.50.0.2
Switch-NEOC	Management	18.50.0.6	255.255.224.0	18.50.0.1	—
MAIL-USR	usr.gov	18.50.32.2	255.255.248.0	18.50.32.1	18.50.0.2
PRN-USR	Printer	18.50.32.3	255.255.248.0	18.50.32.1	18.50.0.2
Switch-USR	Management	18.50.32.4	255.255.248.0	18.50.32.1	—
WEB-MEC	www.mec.alert.gov	18.50.40.2	255.255.254.0	18.50.40.1	18.50.0.2
MAIL-MEC	mec.gov	18.50.40.3	255.255.254.0	18.50.40.1	18.50.0.2
DHCP-MEC	DHCP server	18.50.40.4	255.255.254.0	18.50.40.1	18.50.0.2
PRN-MEC	Printer	18.50.40.5	255.255.254.0	18.50.40.1	18.50.0.2
Switch-MEC	Management	18.50.40.6	255.255.254.0	18.50.40.1	—
MAIL-FRH	frh.gov	18.50.42.2	255.255.254.0	18.50.42.1	18.50.0.2
PRN-FRH	Printer	18.50.42.3	255.255.254.0	18.50.42.1	18.50.0.2
Switch-FRH	Management	18.50.42.4	255.255.254.0	18.50.42.1	—
MAIL-SMS	sms.gov	18.50.44.2	255.255.255.0	18.50.44.1	18.50.0.2
PRN-SMS	Printer	18.50.44.3	255.255.255.0	18.50.44.1	18.50.0.2
Switch-SMS	Management	18.50.44.4	255.255.255.0	18.50.44.1	—
MAIL-CRC	crc.gov	18.50.45.2	255.255.255.128	18.50.45.1	18.50.0.2
DHCP-CRC	DHCP server	18.50.45.3	255.255.255.128	18.50.45.1	18.50.0.2
PRN-CRC	Printer	18.50.45.4	255.255.255.128	18.50.45.1	18.50.0.2
Switch-CRC	Management	18.50.45.5	255.255.255.128	18.50.45.1	—
MAIL-MOP	mop.gov	18.50.45.130	255.255.255.192	18.50.45.129	18.50.0.2
PRN-MOP	Printer	18.50.45.131	255.255.255.192	18.50.45.129	18.50.0.2
Switch-MOP	Management	18.50.45.132	255.255.255.192	18.50.45.129	—
Central-Switch	Management	18.50.45.197	255.255.255.248	18.50.45.193	—

All fourteen PCs (two per unit) obtain their addressing dynamically through DHCP. Every switch management address falls inside a DHCP excluded range (or outside every dynamic pool) so it can never conflict with a dynamically assigned address.

7. DHCP Configuration

DHCP is provided in three ways, as the specification requires: centrally from the NEOC router, from dedicated servers in MEC and CRC, and through DHCP relay for the units with no local DHCP service.

7.1 Centralised DHCP on the NEOC router

The NEOC router serves three pools — its own LAN, SMS and FRH. The first nine addresses of each subnet are excluded so the router interface, servers, printer and switch management address stay reserved.

```
ip dhcp excluded-address 18.50.0.1 18.50.0.9
ip dhcp excluded-address 18.50.44.1 18.50.44.9
ip dhcp excluded-address 18.50.42.1 18.50.42.9
ip dhcp pool NEOC-POOL
 network 18.50.0.0 255.255.224.0
 default-router 18.50.0.1
 dns-server 18.50.0.2
 domain-name neoc.gov
ip dhcp pool SMS-POOL
 network 18.50.44.0 255.255.255.0
 default-router 18.50.44.1
 dns-server 18.50.0.2
 domain-name sms.gov
ip dhcp pool FRH-POOL
 network 18.50.42.0 255.255.254.0
 default-router 18.50.42.1
 dns-server 18.50.0.2
 domain-name frh.gov
```

7.2 DHCP relay on the remote routers

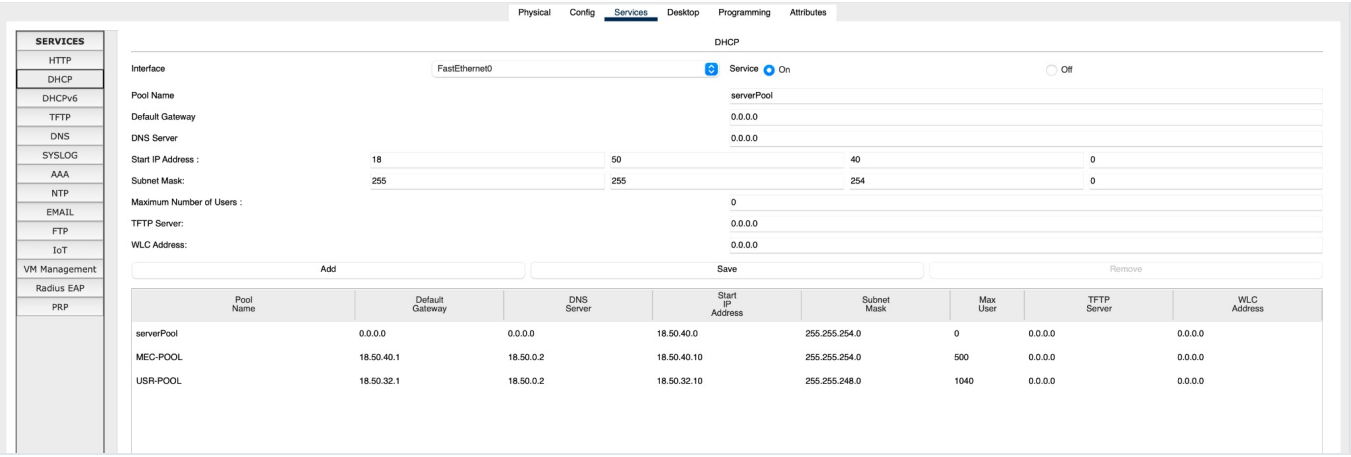
A DHCP discover is a broadcast and a router does not forward it. Four units therefore carry an `ip helper-address` statement on their LAN interface, converting the broadcast into a unicast towards the correct DHCP service.

Router	LAN interface	ip helper-address	Points to
Router-SMS	G0/0/0	18.50.0.1	NEOC router DHCP service
Router-FRH	G0/0/0	18.50.0.1	NEOC router DHCP service
Router-USR	G0/0/0	18.50.40.4	DHCP-MEC server
Router-MOP	G0/0/0	18.50.45.3	DHCP-CRC server

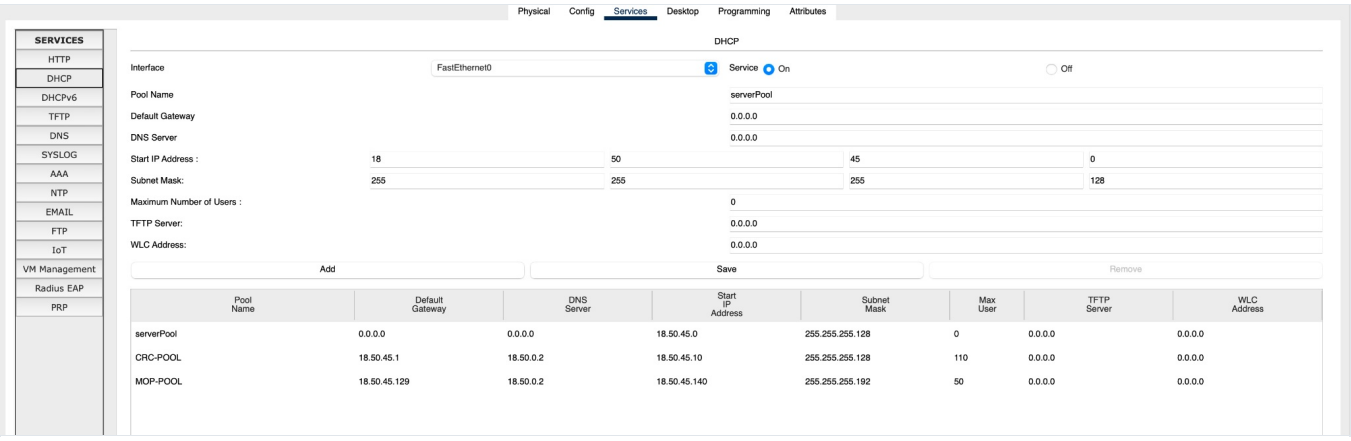
7.3 Dedicated DHCP servers in MEC and CRC

The MEC and CRC units each run a DHCP server (Server-PT). Each holds two pools: one for its own LAN and one for the relayed remote LAN. The default `serverPool` on both servers was disabled so it cannot answer a request in place of the configured pools.

Server	Pool name	Default gateway	DNS server	Start IP	Subnet mask	Max users
DHCP-MEC (18.50.40.4)	MEC-POOL	18.50.40.1	18.50.0.2	18.50.40.10	255.255.254.0	500
DHCP-MEC (18.50.40.4)	USR-POOL	18.50.32.1	18.50.0.2	18.50.32.10	255.255.248.0	1040
DHCP-CRC (18.50.45.3)	CRC-POOL	18.50.45.1	18.50.0.2	18.50.45.10	255.255.255.128	110
DHCP-CRC (18.50.45.3)	MOP-POOL	18.50.45.129	18.50.0.2	18.50.45.140	255.255.255.192	50



DHCP-MEC showing both pools



DHCP-CRC showing both pools



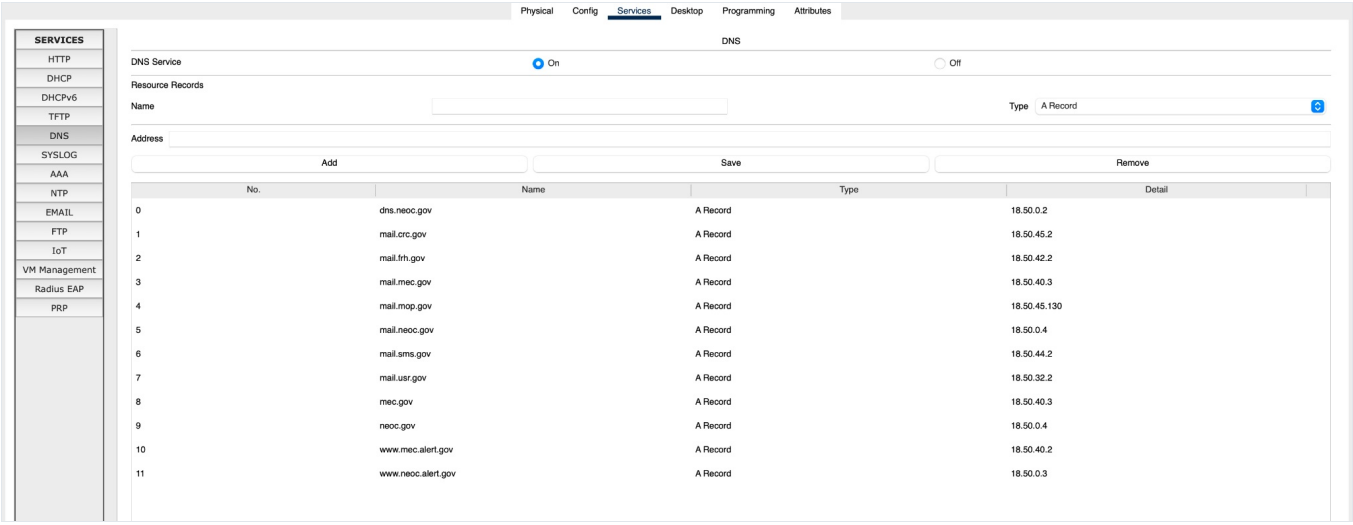
Router-NEOC ip route

8. DNS Configuration

The central DNS server, DNS-NEOC at 18.50.0.2, is the only DNS server in the network and is used by every unit. All servers, printers and DHCP pools point to it.

#	Name	Type	Address
1	www.neoc.alert.gov	A	18.50.0.3
2	www.mec.alert.gov	A	18.50.40.2
3	mail.neoc.gov	A	18.50.0.4
4	mail.mec.gov	A	18.50.40.3
5	mail.sms.gov	A	18.50.44.2
6	mail.frh.gov	A	18.50.42.2
7	mail.usr.gov	A	18.50.32.2
8	mail.crc.gov	A	18.50.45.2
9	mail.mop.gov	A	18.50.45.130
10	neoc.gov	A	18.50.0.4
11	mec.gov	A	18.50.40.3
12	dns.neoc.gov	A	18.50.0.2

Records 10 and 11 are essential. When a mail server relays a message to another domain it resolves the bare domain name, not the `mail.` host name, so without an A record for `neoc.gov` and `mec.gov` the cross-domain delivery would fail even though every other record is correct.



DNS-NEOC showing the complete record list

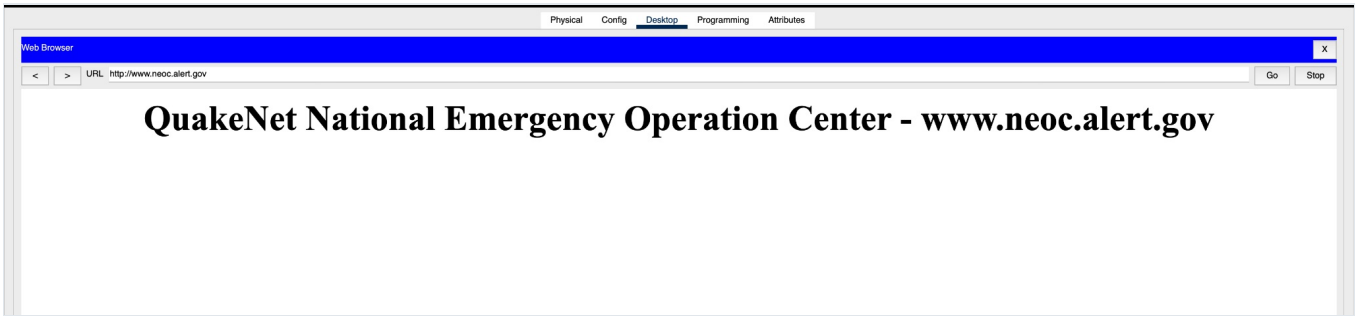
9. Web Servers

Only NEOC and MEC host web servers. Both run HTTP and serve a distinct index page so browser screenshots show which server answered.

Server	IP address	Domain name	Service

WEB-NEOC	18.50.0.3	www.neoc.alert.gov	HTTP enabled
WEB-MEC	18.50.40.2	www.mec.alert.gov	HTTP enabled

Both sites were opened by domain name, not IP address, from a PC in every one of the seven units, which simultaneously proves end-to-end routing, DNS resolution and web server operation.



www.neoc.alert.gov opened from a PC



www.mec.alert.gov opened from a PC

10. Email Servers

Every unit hosts its own email server with the domain `<unit>.gov`. NEOC and MEC carry two accounts each because cross-domain email between those two units with at least two users each is required.

Mail server	IP address	Domain	User accounts	Password
MAIL-NEOC	18.50.0.4	neoc.gov	neoc.admin, neoc.ops	cisco123
MAIL-MEC	18.50.40.3	mec.gov	mec.doctor, mec.nurse	cisco123
MAIL-SMS	18.50.44.2	sms.gov	sms.user1	cisco123
MAIL-FRH	18.50.42.2	frh.gov	frh.user1	cisco123
MAIL-USR	18.50.32.2	usr.gov	usr.user1	cisco123
MAIL-CRC	18.50.45.2	crc.gov	crc.user1	cisco123
MAIL-MOP	18.50.45.130	mop.gov	mop.user1	cisco123

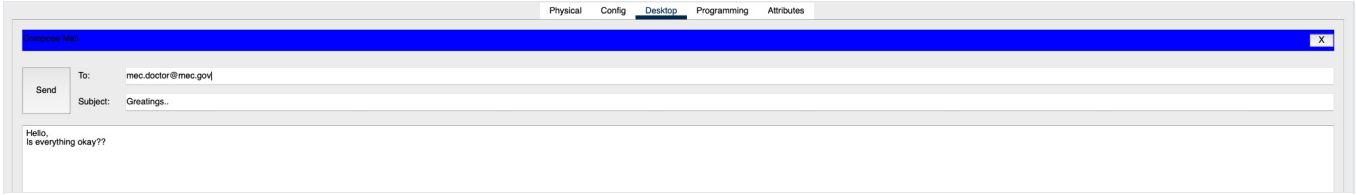
SMTP and POP3 were enabled on all seven servers.

10.1 Mail client configuration on the PCs

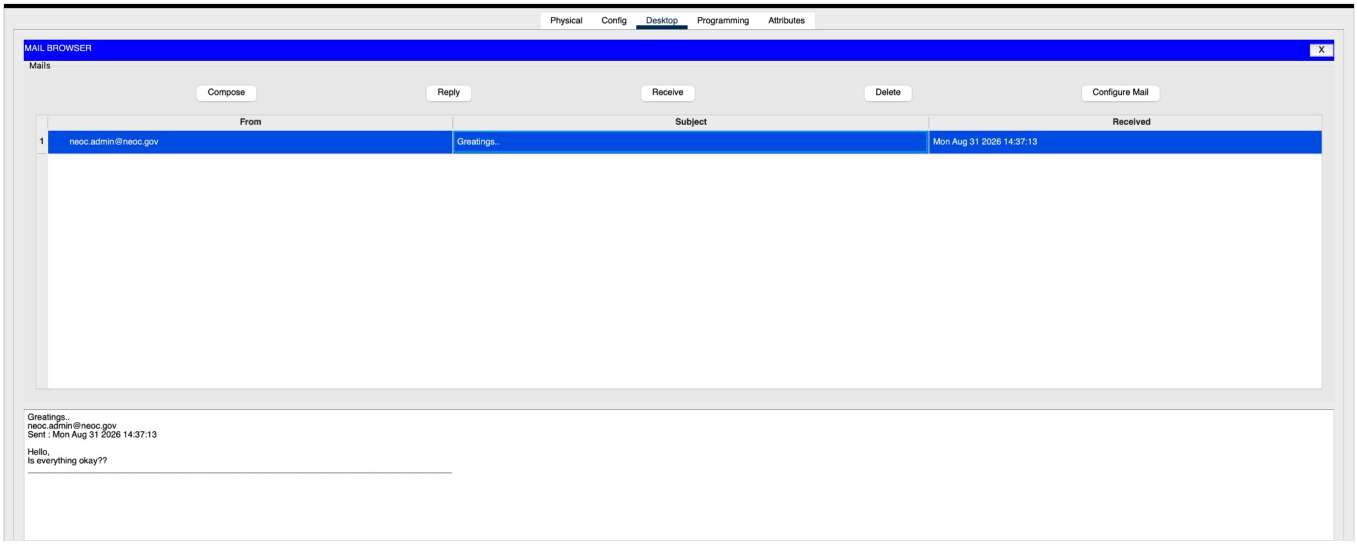
PC	Display name	Email address	Incoming / outgoing server
PC-NEOC-1	NEOC Admin	neoc.admin@neoc.gov	mail.neoc.gov
PC-NEOC-2	NEOC Ops	neoc.ops@neoc.gov	mail.neoc.gov
PC-MEC-1	MEC Doctor	mec.doctor@mec.gov	mail.mec.gov
PC-MEC-2	MEC Nurse	mec.nurse@mec.gov	mail.mec.gov

10.2 Cross-domain mail delivery

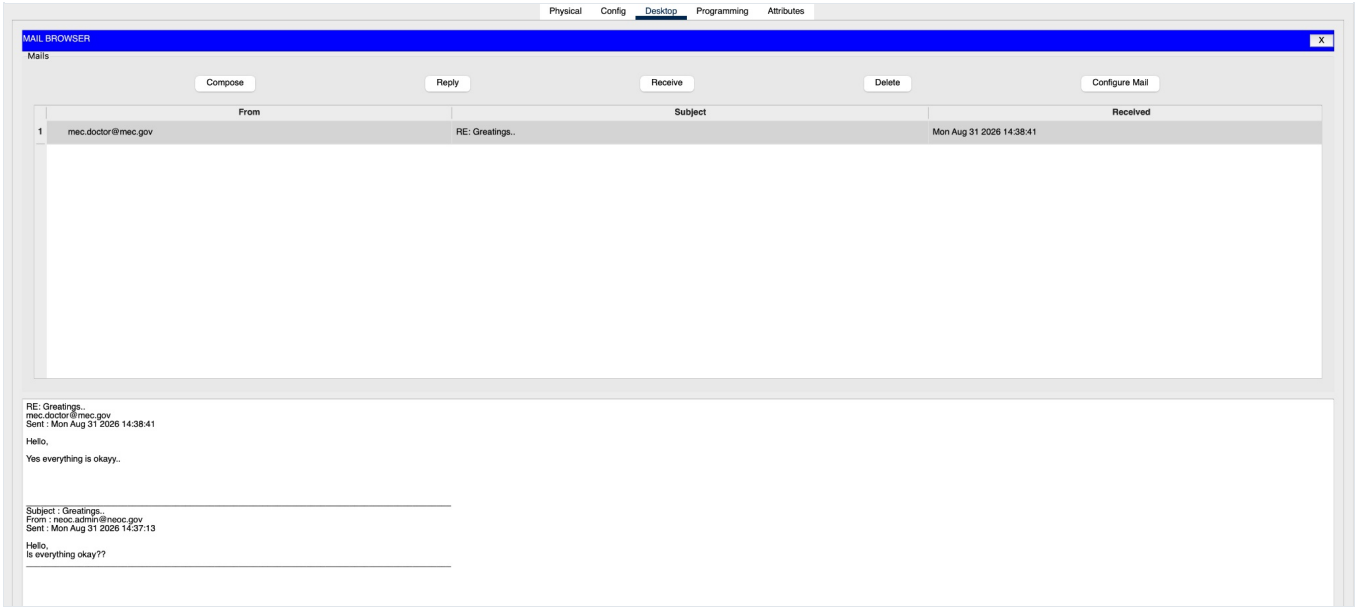
Cross-domain delivery was verified in both directions. A message from neoc.admin@neoc.gov to mec.doctor@mec.gov leaves the NEOC mail server, which resolves `mec.gov` through DNS-NEOC, obtains 18.50.40.3 and relays the message over the fibre link. The reply travels the same path in reverse. The test was repeated for the second user pair, neoc.ops and mec.nurse.



Composing a message from neoc.admin@neoc.gov to mec.doctor@mec.gov



The message received on PC-MEC-1



The reply from mec.doctor@mec.gov received on PC-NEOC-1

11. Routing Configuration

The routing design divides the network into a dynamically routed half and a statically routed half, joined by route redistribution on the NEOC router. No default route exists anywhere.

Router	Routing method	Notes
Router-NEOC	RIPv2 + static + redistribution	RIPv2 on the warning loop; redistributes its static routes with metric 5
Router-SMS	RIPv2 only	Two serial interfaces active in RIP, LAN interface passive
Router-FRH	RIPv2 only	Two serial interfaces active in RIP, LAN and central-switch interfaces passive
Router-MEC	Static only	Recursive next-hop routes with a recursive backup through CRC
Router-CRC	Static only	Recursive next-hop routes with a floating

Router-CRC	Static only	Recursive next-hop routes with a floating static route through FRH, tracked with IP SLA
Router-USR	Static only	Exit-interface (directly attached) routes only
Router-MOP	Static only	Recursive next-hop routes towards CRC

11.1 Dynamic routing — RIPv2

RIPv2 runs on NEOC, SMS and FRH, which form the warning loop. Version 2 is classless and carries subnet masks, which is mandatory for a VLSM design; automatic summarisation is disabled for the same reason.

```
router rip
 version 2
 no auto-summary
 network 18.0.0.0
 passive-interface GigabitEthernet0/0/0
 passive-interface GigabitEthernet0/0/1
 passive-interface GigabitEthernet0/0/2
 redistribute static metric 5
```

`network 18.0.0.0` activates RIP on every interface in the block, so the LAN, central-segment and fibre interfaces are made passive. A passive interface is still advertised in updates but sends none. This keeps RIP traffic away from MEC and CRC (which run static routing only) while their segments stay reachable from the RIP domain. `redistribute static` injects NEOC's static routes into RIPv2, so SMS and FRH learn the MEC, USR, CRC and MOP networks without any static configuration of their own.

11.2 Static routes and redistribution on the NEOC router

```
ip route 18.50.40.0   255.255.254.0   18.50.45.202    ! MEC LAN via fibre
ip route 18.50.32.0   255.255.248.0   18.50.45.202    ! USR LAN via MEC
ip route 18.50.45.216 255.255.255.252 18.50.45.202    ! MEC-USR link
ip route 18.50.45.0   255.255.255.128 18.50.45.196    ! CRC LAN
ip route 18.50.45.128 255.255.255.192 18.50.45.196    ! MOP LAN
ip route 18.50.45.220 255.255.255.252 18.50.45.196    ! CRC-MOP link
```

11.3 Recursive static routes on the MEC router

MEC uses recursive (next-hop) static routes only: the route names the next-hop IP but no exit interface, so the router performs a second lookup to find how to reach that next hop.

```
ip route 18.50.0.0    255.255.224.0   18.50.45.201    ! NEOC LAN over the fibre
ip route 18.50.44.0   255.255.255.0   18.50.45.201    ! SMS LAN
ip route 18.50.42.0   255.255.254.0   18.50.45.194    ! FRH LAN via central switch
ip route 18.50.45.0   255.255.255.128 18.50.45.196    ! CRC LAN
ip route 18.50.45.128 255.255.255.192 18.50.45.196    ! MOP LAN
ip route 18.50.32.0   255.255.248.0   18.50.45.218    ! USR LAN
```

11.4 Static routes on the CRC router

CRC reaches the NEOC and SMS networks through the central switch (NEOC), the FRH networks through the FRH router, the MEC and USR networks through the MEC router, and the MOP LAN through the MOP end of the CRC–MOP link (18.50.45.222). It also carries the floating static route described in Section 12.2.

The route to the MOP LAN (`ip route 18.50.45.128 255.255.255.192 18.50.45.222`) is present in the Packet Tracer routing-table capture (Router-CRC, `S 18.50.45.128/26 [1/0] via 18.50.45.222`) but was missing from the configuration listing printed in the original report. It is restored in `configs/routers/Router-CRC.txt`; the automated verifier shows that without it the MOP LAN is unreachable.

11.5 Exit-interface static routes on the USR router

USR uses directly attached static routes, which name only the exit interface. This is valid because the USR uplink is a point-to-point serial link on which no next-hop resolution is needed. Twelve routes cover every remote network — seven LAN/segment networks and five WAN /30 networks (the sixth, MEC–USR, is directly connected); no default route is used.

```
ip route 18.50.0.0    255.255.224.0   Serial0/1/0
ip route 18.50.40.0   255.255.254.0   Serial0/1/0
ip route 18.50.42.0   255.255.254.0   Serial0/1/0
ip route 18.50.44.0   255.255.255.0   Serial0/1/0
ip route 18.50.45.0   255.255.255.128 Serial0/1/0
ip route 18.50.45.128 255.255.255.192 Serial0/1/0
ip route 18.50.45.192 255.255.255.248 Serial0/1/0
ip route 18.50.45.200 255.255.255.252 Serial0/1/0
ip route 18.50.45.204 255.255.255.252 Serial0/1/0
ip route 18.50.45.208 255.255.255.252 Serial0/1/0
ip route 18.50.45.212 255.255.255.252 Serial0/1/0
ip route 18.50.45.220 255.255.255.252 Serial0/1/0
```

11.6 Static routes on the MOP router

MOP is a pure stub with a single serial uplink to CRC. Because the specification forbids a default route, twelve explicit next-hop static routes towards 18.50.45.221 are configured instead (see `configs/routers/Router-MOP.txt`).

12. Recursive and Floating Backup Routes

Two independent backup mechanisms are implemented, both using administrative distance 48. A route with AD 48 is less preferred than the corresponding primary static route (AD 1), so it stays out of the routing table until the primary route is withdrawn.

12.1 Recursive backup route on the MEC router

MEC reaches the NEOC networks over the point-to-point fibre link as its primary path. A second set of recursive routes points at CRC with administrative distance 48. When the fibre interface goes down, the 18.50.45.200/30 connected route disappears, the recursive lookup for next hop 18.50.45.201 fails, the primary route is removed, and the AD 48 route through CRC installs automatically.

```
! primary (AD 1)
ip route 18.50.0.0    255.255.224.0   18.50.45.201
ip route 18.50.44.0   255.255.255.0   18.50.45.201
ip route 18.50.45.204 255.255.255.252 18.50.45.201
! recursive backup via CRC (AD 48)
ip route 18.50.0.0    255.255.224.0   18.50.45.196 48
ip route 18.50.44.0   255.255.255.0   18.50.45.196 48
ip route 18.50.45.204 255.255.255.252 18.50.45.196 48
```

12.2 Floating static route on the CRC router

CRC maintains a floating static route towards NEOC through FRH. The primary next hop (18.50.45.193) and the floating next hop (18.50.45.194) sit on the same directly connected /29 segment, so a recursive lookup for either always succeeds while CRC's own interface is up. The primary route would therefore never be withdrawn and the floating route could never install on its own.

To make the floating route operational, IP SLA with object tracking is used. An ICMP echo probe monitors the NEOC next hop and the primary routes are bound to that track object. When NEOC stops responding, the track goes down, the primary route is withdrawn, and the AD 48 route through FRH installs.

```
ip sla 1
 icmp-echo 18.50.45.193 source-interface GigabitEthernet0/0/1
 frequency 5
ip sla schedule 1 life forever start-time now
track 1 ip sla 1 reachability
!
ip route 18.50.0.0  255.255.224.0 18.50.45.193 track 1    ! primary, AD 1
ip route 18.50.44.0 255.255.255.0  18.50.45.193 track 1
!
ip route 18.50.0.0  255.255.224.0 18.50.45.194 48         ! floating via FRH
ip route 18.50.44.0 255.255.255.0  18.50.45.194 48
```

12.3 Return-path backup routes on the NEOC router

A backup path is only usable if it exists in both directions. If only MEC had a backup route, traffic would reach NEOC during a fibre failure but the replies would be dropped. NEOC therefore carries matching AD 48 routes towards the MEC and USR networks through CRC.

```
ip route 18.50.40.0   255.255.254.0   18.50.45.196 48
ip route 18.50.32.0   255.255.248.0   18.50.45.196 48
ip route 18.50.45.216 255.255.255.252 18.50.45.196 48
```

Because NEOC redistributes its static routes into RIPv2, these backup routes are propagated to SMS and FRH automatically when they become active.

13. Device Configurations

The complete configurations are included in the repository as text files that can be pasted into each device's CLI (`configs/routers/` and `configs/switches/`). The listings below are generated from those files.

Router-NEOC configuration

```
enable
configure terminal
hostname Router-NEOC
no ip domain-lookup
banner motd #QuakeNet - National Emergency Operation Center - Authorized Access Only#
!
interface GigabitEthernet0/0/0
 description NEOC LAN
 ip address 18.50.0.1 255.255.224.0
 no shutdown
 exit
interface GigabitEthernet0/0/1
 description Central switch segment
 ip address 18.50.45.193 255.255.255.248
 no shutdown
 exit
interface GigabitEthernet0/0/2
 description P2P fibre link to MEC
 ip address 18.50.45.201 255.255.255.252
 no shutdown
 exit
interface Serial0/1/0
 description Warning loop to SMS
 ip address 18.50.45.205 255.255.255.252
 clock rate 64000
 no shutdown
 exit
interface Serial0/1/1
 description Warning loop to FRH
 ip address 18.50.45.214 255.255.255.252
 clock rate 64000
 no shutdown
 exit
!
! ---------- DHCP server for NEOC, SMS and FRH ----------
ip dhcp excluded-address 18.50.0.1 18.50.0.9
ip dhcp excluded-address 18.50.44.1 18.50.44.9
ip dhcp excluded-address 18.50.42.1 18.50.42.9
ip dhcp pool NEOC-POOL
 network 18.50.0.0 255.255.224.0
 default-router 18.50.0.1
 dns-server 18.50.0.2
 domain-name neoc.gov
 exit
ip dhcp pool SMS-POOL
 network 18.50.44.0 255.255.255.0
 default-router 18.50.44.1
 dns-server 18.50.0.2
 domain-name sms.gov
 exit
ip dhcp pool FRH-POOL
 network 18.50.42.0 255.255.254.0
 default-router 18.50.42.1
 dns-server 18.50.0.2
 domain-name frh.gov
 exit
!
! ---------- Static routes toward the static-only side ----------
ip route 18.50.40.0 255.255.254.0 18.50.45.202
ip route 18.50.32.0 255.255.248.0 18.50.45.202
ip route 18.50.45.216 255.255.255.252 18.50.45.202
ip route 18.50.45.0 255.255.255.128 18.50.45.196
ip route 18.50.45.128 255.255.255.192 18.50.45.196
ip route 18.50.45.220 255.255.255.252 18.50.45.196
! Floating backups so the return path also survives a fibre failure
ip route 18.50.40.0 255.255.254.0 18.50.45.196 48
ip route 18.50.32.0 255.255.248.0 18.50.45.196 48
ip route 18.50.45.216 255.255.255.252 18.50.45.196 48
!
! ---------- Dynamic routing on the warning loop ----------
router rip
 version 2
 no auto-summary
 network 18.0.0.0
 passive-interface GigabitEthernet0/0/0
 passive-interface GigabitEthernet0/0/1
 passive-interface GigabitEthernet0/0/2
 redistribute static metric 5
 exit
!
line console 0
 password cisco
 login
 logging synchronous
 exit
enable secret class
service password-encryption
end
copy running-config startup-config
```

Router-SMS configuration

```
enable
configure terminal
hostname Router-SMS
no ip domain-lookup
banner motd #QuakeNet - Seismic Monitoring Station#
!
interface GigabitEthernet0/0/0
 description SMS LAN
 ip address 18.50.44.1 255.255.255.0
 ip helper-address 18.50.0.1
 no shutdown
 exit
interface Serial0/1/0
 description Warning loop to NEOC
 ip address 18.50.45.206 255.255.255.252
 no shutdown
 exit
interface Serial0/1/1
 description Warning loop to FRH
 ip address 18.50.45.209 255.255.255.252
 clock rate 64000
 no shutdown
 exit
!
router rip
 version 2
 no auto-summary
 network 18.0.0.0
 passive-interface GigabitEthernet0/0/0
 exit
!
line console 0
 password cisco
 login
 logging synchronous
 exit
enable secret class
service password-encryption
end
copy running-config startup-config
```

Router-FRH configuration

```
enable
configure terminal
hostname Router-FRH
no ip domain-lookup
banner motd #QuakeNet - Fire and Rescue Headquarters#
!
interface GigabitEthernet0/0/0
 description FRH LAN
 ip address 18.50.42.1 255.255.254.0
 ip helper-address 18.50.0.1
 no shutdown
 exit
interface GigabitEthernet0/0/1
 description Central switch segment
 ip address 18.50.45.194 255.255.255.248
 no shutdown
 exit
interface Serial0/1/0
 description Warning loop to SMS
 ip address 18.50.45.210 255.255.255.252
 no shutdown
 exit
interface Serial0/1/1
 description Warning loop to NEOC
 ip address 18.50.45.213 255.255.255.252
 no shutdown
 exit
!
router rip
 version 2
 no auto-summary
 network 18.0.0.0
 passive-interface GigabitEthernet0/0/0
 passive-interface GigabitEthernet0/0/1
 exit
!
line console 0
 password cisco
 login
 logging synchronous
 exit
enable secret class
service password-encryption
end
copy running-config startup-config
```

Router-MEC configuration

```
enable
configure terminal
hostname Router-MEC
no ip domain-lookup
banner motd #QuakeNet - Medical Emergency Center#
!
interface GigabitEthernet0/0/0
 description MEC LAN
 ip address 18.50.40.1 255.255.254.0
 no shutdown
 exit
interface GigabitEthernet0/0/1
 description Central switch segment
 ip address 18.50.45.195 255.255.255.248
 no shutdown
 exit
interface GigabitEthernet0/0/2
 description P2P fibre link to NEOC
 ip address 18.50.45.202 255.255.255.252
 no shutdown
 exit
interface Serial0/1/0
 description Link to USR
 ip address 18.50.45.217 255.255.255.252
 clock rate 64000
 no shutdown
 exit
!
! ---------- PRIMARY recursive statics over the fibre ----------
ip route 18.50.0.0 255.255.224.0 18.50.45.201
ip route 18.50.44.0 255.255.255.0 18.50.45.201
ip route 18.50.45.204 255.255.255.252 18.50.45.201
!
! ---------- RECURSIVE BACKUP via CRC (AD 48) ----------
ip route 18.50.0.0 255.255.224.0 18.50.45.196 48
ip route 18.50.44.0 255.255.255.0 18.50.45.196 48
ip route 18.50.45.204 255.255.255.252 18.50.45.196 48
!
! ---------- Routes over the central segment ----------
ip route 18.50.42.0 255.255.254.0 18.50.45.194
ip route 18.50.45.208 255.255.255.252 18.50.45.194
ip route 18.50.45.212 255.255.255.252 18.50.45.194
ip route 18.50.45.0 255.255.255.128 18.50.45.196
ip route 18.50.45.128 255.255.255.192 18.50.45.196
ip route 18.50.45.220 255.255.255.252 18.50.45.196
!
! ---------- Route to USR ----------
ip route 18.50.32.0 255.255.248.0 18.50.45.218
!
line console 0
 password cisco
 login
 logging synchronous
 exit
enable secret class
service password-encryption
end
copy running-config startup-config
```

Router-CRC configuration

```
enable
configure terminal
hostname Router-CRC
no ip domain-lookup
banner motd #QuakeNet - Communication Relay Center#
!
interface GigabitEthernet0/0/0
 description CRC LAN
 ip address 18.50.45.1 255.255.255.128
 no shutdown
 exit
interface GigabitEthernet0/0/1
 description Central switch segment
 ip address 18.50.45.196 255.255.255.248
 no shutdown
 exit
interface Serial0/1/0
 description Link to MOP
 ip address 18.50.45.221 255.255.255.252
 clock rate 64000
 no shutdown
 exit
!
! ---------- IP SLA probe + track object (watches NEOC on the central segment) ----------
ip sla 1
 icmp-echo 18.50.45.193 source-interface GigabitEthernet0/0/1
 frequency 5
 exit
ip sla schedule 1 life forever start-time now
track 1 ip sla 1 reachability
!
! ---------- PRIMARY routes (the two NEOC-bound routes are bound to track 1) ----------
ip route 18.50.0.0 255.255.224.0 18.50.45.193 track 1
ip route 18.50.44.0 255.255.255.0 18.50.45.193 track 1
ip route 18.50.45.204 255.255.255.252 18.50.45.193
ip route 18.50.45.212 255.255.255.252 18.50.45.193
ip route 18.50.45.200 255.255.255.252 18.50.45.193
ip route 18.50.42.0 255.255.254.0 18.50.45.194
ip route 18.50.45.208 255.255.255.252 18.50.45.194
ip route 18.50.40.0 255.255.254.0 18.50.45.195
ip route 18.50.32.0 255.255.248.0 18.50.45.195
ip route 18.50.45.216 255.255.255.252 18.50.45.195
! MOP LAN behind the CRC-MOP link (present in the Packet Tracer routing table, missing from the printed report)
ip route 18.50.45.128 255.255.255.192 18.50.45.222
!
! ---------- FLOATING STATIC toward NEOC via FRH (AD 48) ----------
ip route 18.50.0.0 255.255.224.0 18.50.45.194 48
ip route 18.50.44.0 255.255.255.0 18.50.45.194 48
ip route 18.50.45.204 255.255.255.252 18.50.45.194 48
ip route 18.50.45.200 255.255.255.252 18.50.45.194 48
!
line console 0
 password cisco
 login
 logging synchronous
 exit
enable secret class
service password-encryption
end
copy running-config startup-config
```

Router-USR configuration

```
enable
configure terminal
hostname Router-USR
no ip domain-lookup
banner motd #QuakeNet - Urban Search and Rescue Camp#
!
interface GigabitEthernet0/0/0
 description USR LAN
 ip address 18.50.32.1 255.255.248.0
 ip helper-address 18.50.40.4
 no shutdown
 exit
interface Serial0/1/0
 description Link to MEC
 ip address 18.50.45.218 255.255.255.252
 no shutdown
 exit
!
! ---------- Exit-interface static routes (no next-hop IP, no default route) ----------
ip route 18.50.0.0 255.255.224.0 Serial0/1/0
ip route 18.50.40.0 255.255.254.0 Serial0/1/0
ip route 18.50.42.0 255.255.254.0 Serial0/1/0
ip route 18.50.44.0 255.255.255.0 Serial0/1/0
ip route 18.50.45.0 255.255.255.128 Serial0/1/0
ip route 18.50.45.128 255.255.255.192 Serial0/1/0
ip route 18.50.45.192 255.255.255.248 Serial0/1/0
ip route 18.50.45.200 255.255.255.252 Serial0/1/0
ip route 18.50.45.204 255.255.255.252 Serial0/1/0
ip route 18.50.45.208 255.255.255.252 Serial0/1/0
ip route 18.50.45.212 255.255.255.252 Serial0/1/0
ip route 18.50.45.220 255.255.255.252 Serial0/1/0
!
line console 0
 password cisco
 login
 logging synchronous
 exit
enable secret class
service password-encryption
end
copy running-config startup-config
```

Router-MOP configuration

```
enable
configure terminal
hostname Router-MOP
no ip domain-lookup
banner motd #QuakeNet - Mountain Observation Post#
!
interface GigabitEthernet0/0/0
 description MOP LAN
 ip address 18.50.45.129 255.255.255.192
 ip helper-address 18.50.45.3
 no shutdown
 exit
interface Serial0/1/0
 description Link to CRC
 ip address 18.50.45.222 255.255.255.252
 no shutdown
 exit
!
ip route 18.50.0.0 255.255.224.0 18.50.45.221
ip route 18.50.32.0 255.255.248.0 18.50.45.221
ip route 18.50.40.0 255.255.254.0 18.50.45.221
ip route 18.50.42.0 255.255.254.0 18.50.45.221
ip route 18.50.44.0 255.255.255.0 18.50.45.221
ip route 18.50.45.0 255.255.255.128 18.50.45.221
ip route 18.50.45.192 255.255.255.248 18.50.45.221
ip route 18.50.45.200 255.255.255.252 18.50.45.221
ip route 18.50.45.204 255.255.255.252 18.50.45.221
ip route 18.50.45.208 255.255.255.252 18.50.45.221
ip route 18.50.45.212 255.255.255.252 18.50.45.221
ip route 18.50.45.216 255.255.255.252 18.50.45.221
!
line console 0
 password cisco
 login
 logging synchronous
 exit
enable secret class
service password-encryption
end
copy running-config startup-config
```

13.1 Switch configuration

All eight switches are configured identically apart from hostname, management address and default gateway. Switch-NEOC is shown in full; the others use the values below.

Switch	VLAN 1 IP address	Subnet mask	Default gateway
Switch-SMS	18.50.44.4	255.255.255.0	18.50.44.1

Switch-FRH	18.50.42.4	255.255.254.0	18.50.42.1
Switch-MEC	18.50.40.6	255.255.254.0	18.50.40.1
Switch-USR	18.50.32.4	255.255.248.0	18.50.32.1
Switch-CRC	18.50.45.5	255.255.255.128	18.50.45.1
Switch-MOP	18.50.45.132	255.255.255.192	18.50.45.129
Central-Switch	18.50.45.197	255.255.255.248	18.50.45.193

Switch-NEOC configuration

```
enable
configure terminal
hostname Switch-NEOC
no ip domain-lookup
interface vlan 1
 ip address 18.50.0.6 255.255.224.0
 no shutdown
 exit
ip default-gateway 18.50.0.1
line console 0
 password cisco
 login
 logging synchronous
 exit
enable secret class
service password-encryption
end
copy running-config startup-config
```

14. Testing and Results

The network was verified in four stages: interface and routing verification, end-to-end connectivity, service verification and failover testing.

14.1 Interface and routing verification

```
show ip interface brief   ! all configured interfaces up / up
show ip route             ! executed on every router
show ip protocols         ! RIPv2, no auto-summary (NEOC, SMS, FRH)
show ip route rip         ! executed on SMS and FRH
show ip dhcp binding      ! executed on Router-NEOC
show cdp neighbors
```

On every router the output confirms that no gateway of last resort is set, satisfying the requirement that no default route exists anywhere in the system.

Router-NEOC

```
QuakeNet - National Emergency Operation Center - Authorized Access Only

User Access Verification

Password:

Router-NEOC>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     18.0.0.0/8 is variably subnetted, 19 subnets, 9 masks
C       18.50.0.0/19 is directly connected, GigabitEthernet0/0/0
L       18.50.0.1/32 is directly connected, GigabitEthernet0/0/0
S       18.50.32.0/21 [1/0] via 18.50.45.202
S       18.50.40.0/23 [1/0] via 18.50.45.202
R       18.50.42.0/23 [120/1] via 18.50.45.213, 00:00:29, Serial0/1/1
R       18.50.44.0/24 [120/1] via 18.50.45.206, 00:00:10, Serial0/1/0
S       18.50.45.0/25 [1/0] via 18.50.45.196
S       18.50.45.128/26 [1/0] via 18.50.45.196
C       18.50.45.192/29 is directly connected, GigabitEthernet0/0/1
L       18.50.45.193/32 is directly connected, GigabitEthernet0/0/1
C       18.50.45.200/30 is directly connected, GigabitEthernet0/0/2
L       18.50.45.201/32 is directly connected, GigabitEthernet0/0/2
C       18.50.45.204/30 is directly connected, Serial0/1/0
L       18.50.45.205/32 is directly connected, Serial0/1/0
R       18.50.45.208/30 [120/1] via 18.50.45.206, 00:00:10, Serial0/1/0
                        [120/1] via 18.50.45.213, 00:00:29, Serial0/1/1
C       18.50.45.212/30 is directly connected, Serial0/1/1
L       18.50.45.214/32 is directly connected, Serial0/1/1
S       18.50.45.216/30 [1/0] via 18.50.45.202
S       18.50.45.220/30 [1/0] via 18.50.45.196

Router-NEOC>
Router-NEOC>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   18.50.0.1       YES manual up                    up
GigabitEthernet0/0/1   18.50.45.193    YES manual up                    up
GigabitEthernet0/0/2   18.50.45.201    YES manual up                    up
Serial0/1/0            18.50.45.205    YES manual up                    up
Serial0/1/1            18.50.45.214    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router-NEOC>
Router-NEOC>
Router-NEOC>
```

show ip route on Router-NEOC

Router-SMS

```
Router-SMS>
Router-SMS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     18.0.0.0/8 is variably subnetted, 17 subnets, 9 masks
R       18.50.0.0/19 [120/1] via 18.50.45.205, 00:00:18, Serial0/1/0
R       18.50.32.0/21 [120/5] via 18.50.45.205, 00:00:18, Serial0/1/0
R       18.50.40.0/23 [120/5] via 18.50.45.205, 00:00:18, Serial0/1/0
R       18.50.42.0/23 [120/1] via 18.50.45.210, 00:00:05, Serial0/1/1
C       18.50.44.0/24 is directly connected, GigabitEthernet0/0/0
L       18.50.44.1/32 is directly connected, GigabitEthernet0/0/0
R       18.50.45.0/25 [120/5] via 18.50.45.205, 00:00:18, Serial0/1/0
R       18.50.45.128/26 [120/5] via 18.50.45.205, 00:00:18, Serial0/1/0
R       18.50.45.192/29 [120/1] via 18.50.45.210, 00:00:05, Serial0/1/1
                        [120/1] via 18.50.45.205, 00:00:18, Serial0/1/0
R       18.50.45.200/30 [120/1] via 18.50.45.205, 00:00:18, Serial0/1/0
C       18.50.45.204/30 is directly connected, Serial0/1/0
L       18.50.45.206/32 is directly connected, Serial0/1/0
C       18.50.45.208/30 is directly connected, Serial0/1/1
L       18.50.45.209/32 is directly connected, Serial0/1/1
R       18.50.45.212/30 [120/1] via 18.50.45.210, 00:00:05, Serial0/1/1
                        [120/1] via 18.50.45.205, 00:00:18, Serial0/1/0
R       18.50.45.216/30 [120/5] via 18.50.45.205, 00:00:18, Serial0/1/0
R       18.50.45.220/30 [120/5] via 18.50.45.205, 00:00:18, Serial0/1/0

Router-SMS>
Router-SMS>
```

show ip route on Router-SMS

Router-FRH

```
Router-FRH>show ip route
                ^
% Invalid input detected at '^' marker.

Router-FRH>
Router-FRH>
Router-FRH>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     18.0.0.0/8 is variably subnetted, 18 subnets, 9 masks
R       18.50.0.0/19 [120/1] via 18.50.45.214, 00:00:08, Serial0/1/1
R       18.50.32.0/21 [120/5] via 18.50.45.214, 00:00:08, Serial0/1/1
R       18.50.40.0/23 [120/5] via 18.50.45.214, 00:00:08, Serial0/1/1
C       18.50.42.0/23 is directly connected, GigabitEthernet0/0/0
L       18.50.42.1/32 is directly connected, GigabitEthernet0/0/0
R       18.50.44.0/24 [120/1] via 18.50.45.209, 00:00:08, Serial0/1/0
R       18.50.45.0/25 [120/5] via 18.50.45.214, 00:00:08, Serial0/1/1
R       18.50.45.128/26 [120/5] via 18.50.45.214, 00:00:08, Serial0/1/1
C       18.50.45.192/29 is directly connected, GigabitEthernet0/0/1
L       18.50.45.194/32 is directly connected, GigabitEthernet0/0/1
R       18.50.45.200/30 [120/1] via 18.50.45.214, 00:00:08, Serial0/1/1
R       18.50.45.204/30 [120/1] via 18.50.45.214, 00:00:08, Serial0/1/1
                        [120/1] via 18.50.45.209, 00:00:08, Serial0/1/0
C       18.50.45.208/30 is directly connected, Serial0/1/0
L       18.50.45.210/32 is directly connected, Serial0/1/0
C       18.50.45.212/30 is directly connected, Serial0/1/1
L       18.50.45.213/32 is directly connected, Serial0/1/1
R       18.50.45.216/30 [120/5] via 18.50.45.214, 00:00:08, Serial0/1/1
R       18.50.45.220/30 [120/5] via 18.50.45.214, 00:00:08, Serial0/1/1

Router-FRH>
Router-FRH>
```

show ip route on Router-FRH

Router-MEC

```
Router-MEC>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     18.0.0.0/8 is variably subnetted, 18 subnets, 9 masks
S       18.50.0.0/19 [1/0] via 18.50.45.201
S       18.50.32.0/21 [1/0] via 18.50.45.218
C       18.50.40.0/23 is directly connected, GigabitEthernet0/0/0
L       18.50.40.1/32 is directly connected, GigabitEthernet0/0/0
S       18.50.42.0/23 [1/0] via 18.50.45.194
S       18.50.44.0/24 [1/0] via 18.50.45.201
S       18.50.45.0/25 [1/0] via 18.50.45.196
S       18.50.45.128/26 [1/0] via 18.50.45.196
C       18.50.45.192/29 is directly connected, GigabitEthernet0/0/1
L       18.50.45.195/32 is directly connected, GigabitEthernet0/0/1
C       18.50.45.200/30 is directly connected, GigabitEthernet0/0/2
L       18.50.45.202/32 is directly connected, GigabitEthernet0/0/2
S       18.50.45.204/30 [1/0] via 18.50.45.201
S       18.50.45.208/30 [1/0] via 18.50.45.194
S       18.50.45.212/30 [1/0] via 18.50.45.194
C       18.50.45.216/30 is directly connected, Serial0/1/0
L       18.50.45.217/32 is directly connected, Serial0/1/0
S       18.50.45.220/30 [1/0] via 18.50.45.196

Router-MEC>
Router-MEC>
```

show ip route on Router-MEC

Router-CRC

```
Press RETURN to get started!

QuakeNet - Communication Relay Center

User Access Verification

Password:

Router-CRC>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     18.0.0.0/8 is variably subnetted, 17 subnets, 9 masks
S       18.50.0.0/19 [1/0] via 18.50.45.193
S       18.50.32.0/21 [1/0] via 18.50.45.195
S       18.50.40.0/23 [1/0] via 18.50.45.195
S       18.50.42.0/23 [1/0] via 18.50.45.194
S       18.50.44.0/24 [1/0] via 18.50.45.193
C       18.50.45.0/25 is directly connected, GigabitEthernet0/0/0
L       18.50.45.1/32 is directly connected, GigabitEthernet0/0/0
S       18.50.45.128/26 [1/0] via 18.50.45.222
C       18.50.45.192/29 is directly connected, GigabitEthernet0/0/1
L       18.50.45.196/32 is directly connected, GigabitEthernet0/0/1
S       18.50.45.200/30 [1/0] via 18.50.45.193
S       18.50.45.204/30 [1/0] via 18.50.45.193
S       18.50.45.208/30 [1/0] via 18.50.45.194
S       18.50.45.212/30 [1/0] via 18.50.45.193
S       18.50.45.216/30 [1/0] via 18.50.45.195
C       18.50.45.220/30 is directly connected, Serial0/1/0
L       18.50.45.221/32 is directly connected, Serial0/1/0

Router-CRC>
Router-CRC>
```

show ip route on Router-CRC

Router-USR

```
User Access Verification

Password:

Router-USR>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     18.0.0.0/8 is variably subnetted, 16 subnets, 9 masks
S       18.50.0.0/19 is directly connected, Serial0/1/0
C       18.50.32.0/21 is directly connected, GigabitEthernet0/0/0
L       18.50.32.1/32 is directly connected, GigabitEthernet0/0/0
S       18.50.40.0/23 is directly connected, Serial0/1/0
S       18.50.42.0/23 is directly connected, Serial0/1/0
S       18.50.44.0/24 is directly connected, Serial0/1/0
S       18.50.45.0/25 is directly connected, Serial0/1/0
S       18.50.45.128/26 is directly connected, Serial0/1/0
S       18.50.45.192/29 is directly connected, Serial0/1/0
S       18.50.45.200/30 is directly connected, Serial0/1/0
S       18.50.45.204/30 is directly connected, Serial0/1/0
S       18.50.45.208/30 is directly connected, Serial0/1/0
S       18.50.45.212/30 is directly connected, Serial0/1/0
C       18.50.45.216/30 is directly connected, Serial0/1/0
L       18.50.45.218/32 is directly connected, Serial0/1/0
S       18.50.45.220/30 is directly connected, Serial0/1/0

Router-USR>
Router-USR>
```

show ip route on Router-USR

Router-MOP

```
QuakeNet - Mountain Observation Post

User Access Verification

Password:

Router-MOP>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     18.0.0.0/8 is variably subnetted, 16 subnets, 9 masks
S       18.50.0.0/19 [1/0] via 18.50.45.221
S       18.50.32.0/21 [1/0] via 18.50.45.221
S       18.50.40.0/23 [1/0] via 18.50.45.221
S       18.50.42.0/23 [1/0] via 18.50.45.221
S       18.50.44.0/24 [1/0] via 18.50.45.221
S       18.50.45.0/25 [1/0] via 18.50.45.221
C       18.50.45.128/26 is directly connected, GigabitEthernet0/0/0
L       18.50.45.129/32 is directly connected, GigabitEthernet0/0/0
S       18.50.45.192/29 [1/0] via 18.50.45.221
S       18.50.45.200/30 [1/0] via 18.50.45.221
S       18.50.45.204/30 [1/0] via 18.50.45.221
S       18.50.45.208/30 [1/0] via 18.50.45.221
S       18.50.45.212/30 [1/0] via 18.50.45.221
S       18.50.45.216/30 [1/0] via 18.50.45.221
C       18.50.45.220/30 is directly connected, Serial0/1/0
L       18.50.45.222/32 is directly connected, Serial0/1/0

Router-MOP>
Router-MOP>
Router-MOP>
Router-MOP>
```

show ip route on Router-MOP

14.2 Connectivity tests

#	Test	Source	Destination	Purpose	Result
1	ping	PC-MOP-1	18.50.0.3	MOP reaches the NEOC web server through CRC and the central switch	Success
2	ping	PC-USR-1	PC-SMS-1	Exit-interface static routes plus redistribution	Success
3	ping	PC-SMS-1	18.50.45.130	RIP half of the	Success

				network reaches the static half	
4	ping	PC-CRC-1	PC-USR-1	CRC reaches USR through MEC	Success
5	ping	PC-FRH-1	18.50.40.2	FRH reaches the MEC web server	Success
6	tracert	PC-FRH-1	18.50.40.2	FRH → NEOC → MEC forward; MEC's reply uses the central segment (route to the FRH LAN via 18.50.45.194)	Success (asymmetric path)
7	tracert	PC-MOP-1	18.50.0.3	Path must be MOP → CRC → NEOC	Path via CRC
8	ipconfig /all	One PC per unit	—	Confirms dynamic addressing and relay	Lease received
9	browser	One PC per unit	both websites	Confirms DNS and HTTP from all seven units	Both sites open
10	email	PC-NEOC-1	PC-MEC-1	Cross-domain delivery in both directions	Delivered

15. Failover Testing

The two backup mechanisms were tested separately. In each test the primary link was administratively shut down, the routing table was re-examined, the new path was verified with a traceroute, and the link was then restored.

15.1 Test A — recursive backup route, failure of the NEOC ↔ MEC fibre link

1. On Router-MEC run `show ip route 18.50.0.0` and record the primary route via 18.50.45.201 with administrative distance 1.
2. Record the normal path with `tracert 18.50.0.3` from PC-MEC-1; the traffic crosses the fibre link.
3. On Router-NEOC enter interface configuration mode for GigabitEthernet0/0/2 and issue `shutdown`.
4. Wait about ten seconds and run `show ip route 18.50.0.0` on Router-MEC again.
5. The route via 18.50.45.201 is removed and the recursive backup via 18.50.45.196 (AD 48) is installed.
6. Run `ping 18.50.0.3` and `tracert 18.50.0.3` from PC-MEC-1; the traffic now travels through the CRC router.
7. Issue `no shutdown` on GigabitEthernet0/0/2; the original path is restored.

15.2 Test B — floating static route, failure of the NEOC central-switch port

1. On Router-CRC run `show ip route 18.50.0.0` and `show track 1`; record the primary route via 18.50.45.193 and track state Up.
2. Record the normal path with `tracert 18.50.0.3` from PC-CRC-1.
3. On Router-NEOC enter interface configuration mode for GigabitEthernet0/0/1 and issue `shutdown`.
4. Wait about ten seconds and run `show track 1` on Router-CRC; the reachability state is now Down.
5. Run `show ip route 18.50.0.0` on Router-CRC; the floating static route via 18.50.45.194 (AD 48) is now installed.
6. Run `tracert 18.50.0.3` from PC-CRC-1; the forward traffic now travels through the FRH router.
7. Issue `no shutdown` on GigabitEthernet0/0/1; the original path is restored.

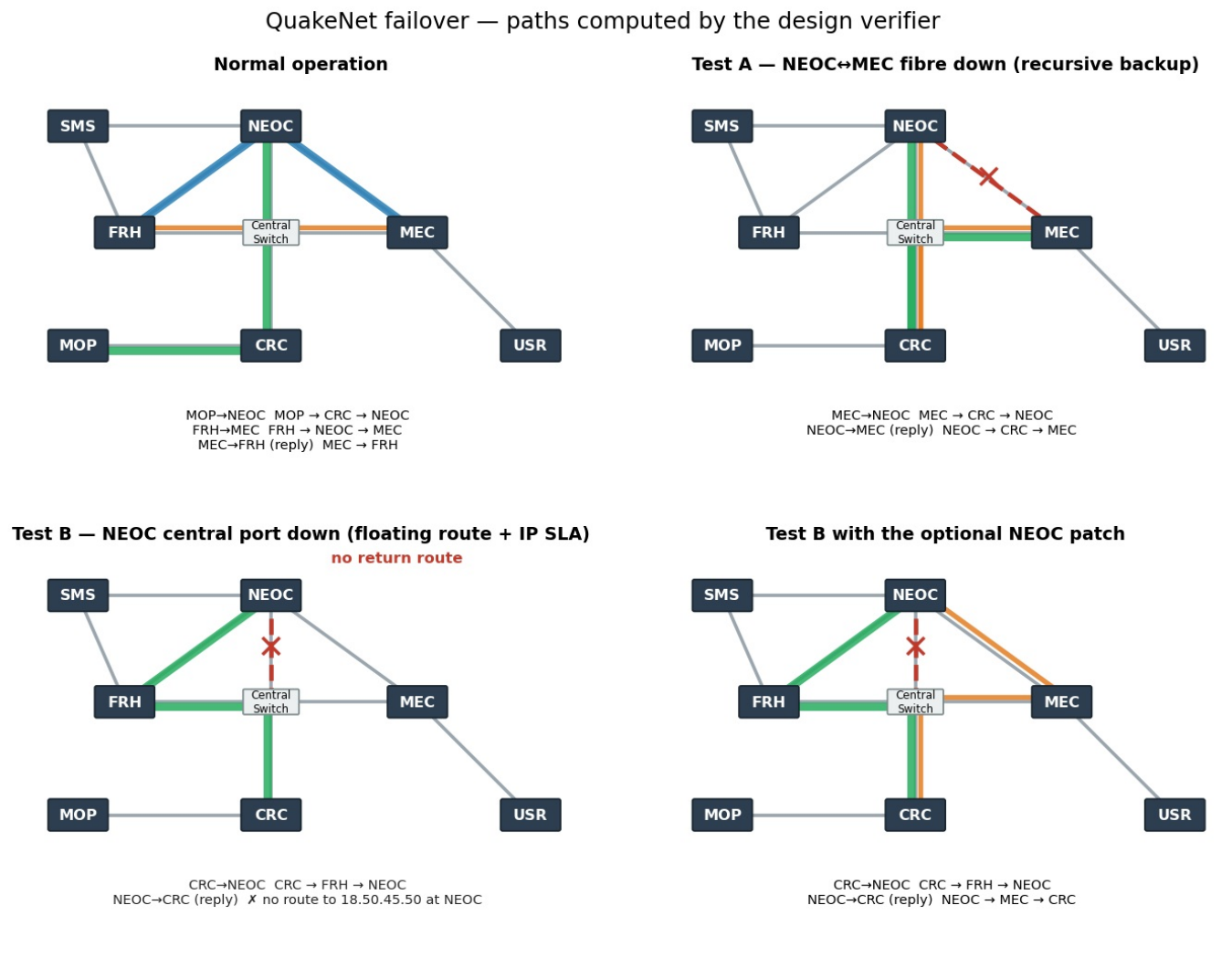
The two tests were run separately; both links were never shut down at the same time.

15.3 Independent verification with the design verifier

In addition to the Packet Tracer tests, the repository contains a Python verifier (`python -m tools.validate_design`) that parses the router configurations, rebuilds each router's routing table (connected, static, floating, tracked, RIPv2 with `redistribute static`) and forwards traffic hop by hop in normal operation and under both failure scenarios. It models the IP SLA probe as reachability of the target on the CRC's central-segment interface. In every state it checks 112 flows: every LAN pair plus every LAN to DNS, both web servers and all seven mail servers. As a calibration, the number of routes the model builds on each of the seven routers equals the `N subnets` count printed in the corresponding Packet Tracer `show ip route` capture (19, 18, 17, 18, 17, 16 and 16 for NEOC, FRH, SMS, MEC, CRC, USR and MOP).

State	Flow	Computed path
Normal	MOP → NEOC	MOP → CRC → NEOC

Normal	FRH → MEC	FRH → NEOC → MEC
Normal	MEC → FRH (reply)	MEC → FRH (central segment)
Test A (fibre down)	MEC → NEOC	MEC → CRC → NEOC
Test A	NEOC → MEC (return)	NEOC → CRC → MEC
Test A	USR → NEOC	USR → MEC → CRC → NEOC
Test B (NEOC central port down)	CRC → NEOC	CRC → FRH → NEOC
Test B	MOP → NEOC	MOP → CRC → FRH → NEOC



Computed failover paths

Normal operation and Test A deliver all 112 flows. The verifier also confirms the design justifications: without NEOC's return-path floating routes Test A breaks the reverse direction; without IP SLA tracking CRC keeps its dead primary route in Test B; and without CRC's route to the MOP LAN, MOP is unreachable.

15.4 Findings

Finding 1 — CRC route to the MOP LAN. The printed CRC listing lacked `ip route 18.50.45.128 255.255.255.192 18.50.45.222`, although the Packet Tracer capture shows it. The route is restored in the repository configuration.

Finding 2 — return traffic in Test B. The forward failover works (CRC and MOP reach NEOC through FRH once track 1 goes down), but NEOC's own static routes to the CRC and MOP LANs use next hop 18.50.45.196 on the central segment. While NEOC's central port is shut those routes cannot resolve and nothing replaces them, so NEOC (and, through redistribution, SMS and FRH) has no route back to the CRC and MOP LANs: 12 of 112 flows fail in the model, all towards those LANs. The three-line optional patch `configs/patches/NEOC_testB_return_path.txt` adds floating routes on NEOC via the fibre link to MEC (which already routes to CRC); in the model it restores all 112 flows in Test B and leaves NEOC's routing table unchanged in normal operation and in Test A. It is *not* applied in the Packet Tracer file, and the finding comes from the verifier's model, not from a Packet Tracer run.

Finding 3 — asymmetric path FRH ↔ MEC. FRH reaches the MEC LAN through NEOC (RIP route from NEOC's redistributed static), while MEC reaches the FRH LAN directly over the central segment. Connectivity test 6 was therefore reworded to describe the forward and reply paths separately.

16. Conclusion and Limitations

QuakeNet implements a complete national earthquake early-warning and emergency-response infrastructure for seven operational units within a single /16 block. VLSM kept the addressing efficient, consuming only 18.50.0.0 through 18.50.45.223 and leaving the remainder free for future units.

Dynamic and static routing coexist in one network: RIPv2 handles the three routers of the warning loop, while the four remaining units are reached through static routes in different styles (recursive next-hop and directly attached). Redistribution lets the dynamically routed part learn the statically routed part without manual configuration on SMS and FRH.

The backup routes were the most instructive part. The recursive backup on MEC fails over on its own, because the fibre is a physically separate interface: losing it removes the connected network on which the primary next hop depends. The floating route on CRC behaves differently — both candidate next hops lie on the same connected segment, so the primary never becomes invalid by itself and IP SLA object tracking was needed to make the failover observable. A floating static route is only as reliable as the mechanism that withdraws the route it backs up. Backups were also needed in both directions, which is why NEOC carries return-path routes; Finding 2 shows the same principle applies to the central-port failure.

Limitations of the design:

- **Single core router.** NEOC hosts the central DNS, a web server, a mail server and DHCP for three units; its failure disables name resolution for the whole system. A second core router with HSRP would address this.
- **MOP and USR have a single uplink**, so neither has any backup path.
- **Test B return path.** NEOC has no route back to the CRC and MOP LANs while its central-switch port is down (Finding 2); an optional patch is provided but not applied.
- **RIPv2** has a 15-hop limit and slow convergence; OSPF would scale better and remove the manual redistribution.
- **Static routing** on MEC, CRC, USR and MOP does not scale: every new subnet needs a manual route statement on four routers.
- **No MX records.** Packet Tracer does not implement them, so cross-domain email relies on A records for the mail domains.
- **Security** features (VLAN segmentation, ACLs, port security, SSH management) were outside the scope of this project.